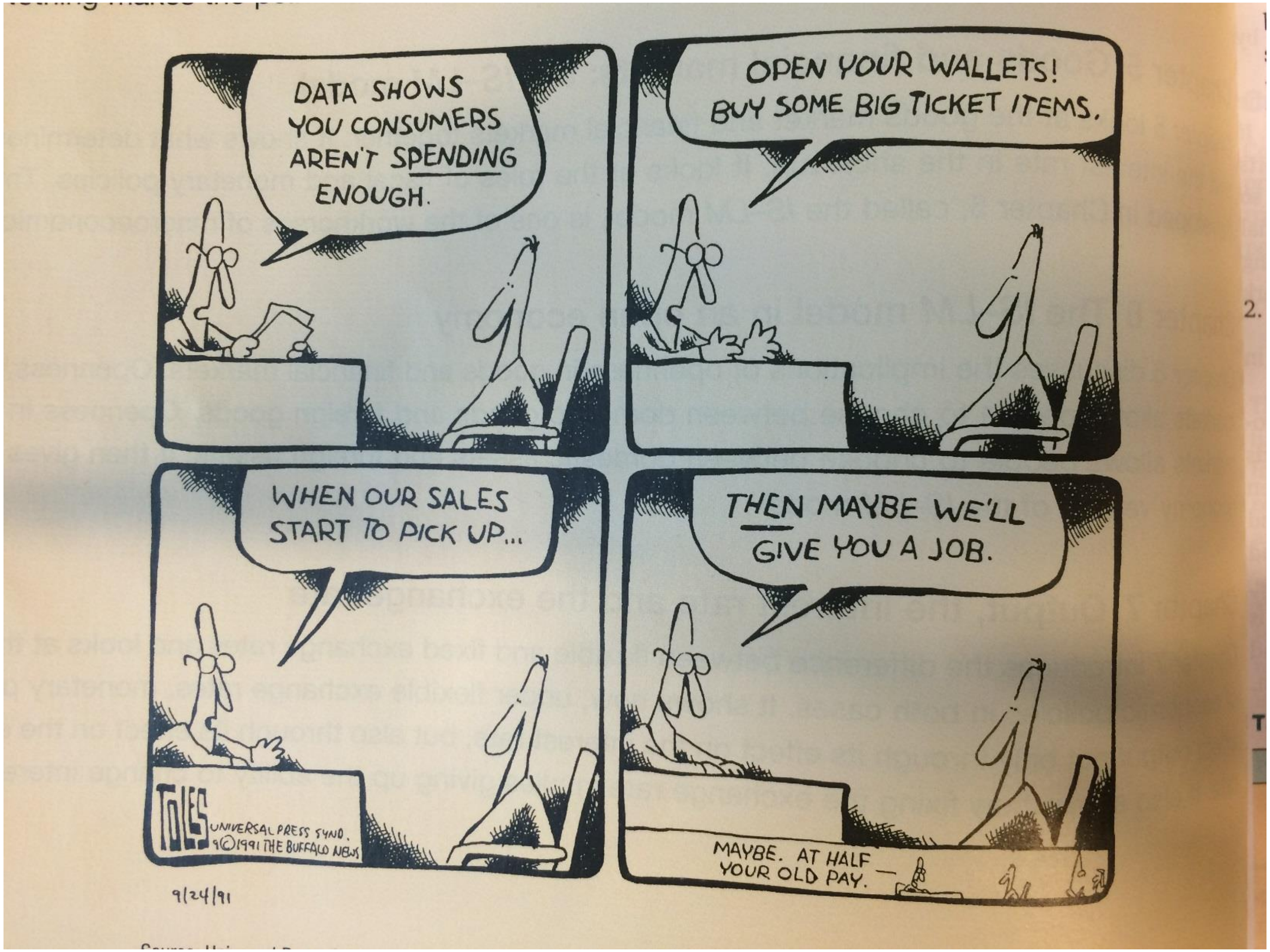
Chapter 3
The goods market

Demand → Production → Income → Demand →…



Composition of GDP

in terms of the different goods produced and the different buyers of the goods

EU15, 2010

	Billion euros	Percentage of GDP
GDP (*Y*)	11,634	100
1 Consumption (*C*)	6,735	57.9
2 Investment (*I*)	1,892	16.3
3 Government spending (*G*)	2,566	22.1
4 Net exports	136	1.1
Exports (*X*)	4,890	42
Imports (*IM*)	–4,754	–40.9
5 Inventory investment	60	0.5

- **Consumption (C):**
    - purchases of goods and services by consumers
    - sum of durables (cars) and non-durable consumption (food)

- **Investment (I):**
    - purchases of capital goods
    - sum of residential investment (housing by people) and non-residential investment (plants and machines by firms)

- **Government spending (G):**
  - purchases of goods and services by the government
  - does not include government transfers, nor interest payments on government debt
  - different from total government spending

- **Imports (IM):** purchases of foreign goods and services by domestic consumers, business firms, and the government

- **Exports (X):** purchases of domestic goods and services by foreigners

- **Net exports or trade balance (X – IM):** difference between exports and imports, also called the trade balance

Exports > imports → *trade surplus*

Exports < imports → *trade deficits*

- **Inventory investment:** difference between production and sales (some of the goods produced in a given year are not sold in the same year, but in later years; some of the goods sold in a given year have been produced in earlier years)

The demand of goods

The total demand for goods is given by

$$Z \equiv C + I + G + X - IM$$

This is an identity equation or a definition

To determine Z and Y, we make 3 simplifying assumptions:

1. Assume that all firms produce the **same good**, which can then be used by consumers for consumption, by firms for investment or by the government → we can look at only one market

2. Assume that firms are willing to supply **any amount** of the good at **given price *P*** and meet the demand in that market → we focus on the role of demand and disregard (for now) the role of supply

3. Assume that the economy is **closed**, that it does not trade with the rest of the world, then both exports and imports are zero, that is, X=IM=0 → we disregard (for now) the role of trade

Consumption (C)

- **Disposable income (Y_D):**

$$Y_D \equiv Y - T$$

  - income that remains once consumers have paid taxes and received transfers from the government
  - Y is gross income; T is taxes net of transfers

- **Consumption function:**

$$C = C(\underset{(+)}{Y_D})$$

  - behavioral equation capturing the behavior of consumer
  - consumption depends positively on disposable income

Example, linear consumption function:

$$C = c_0 + c_1 Y_D$$

This function has **2 parameters**, c_0 and c_1:

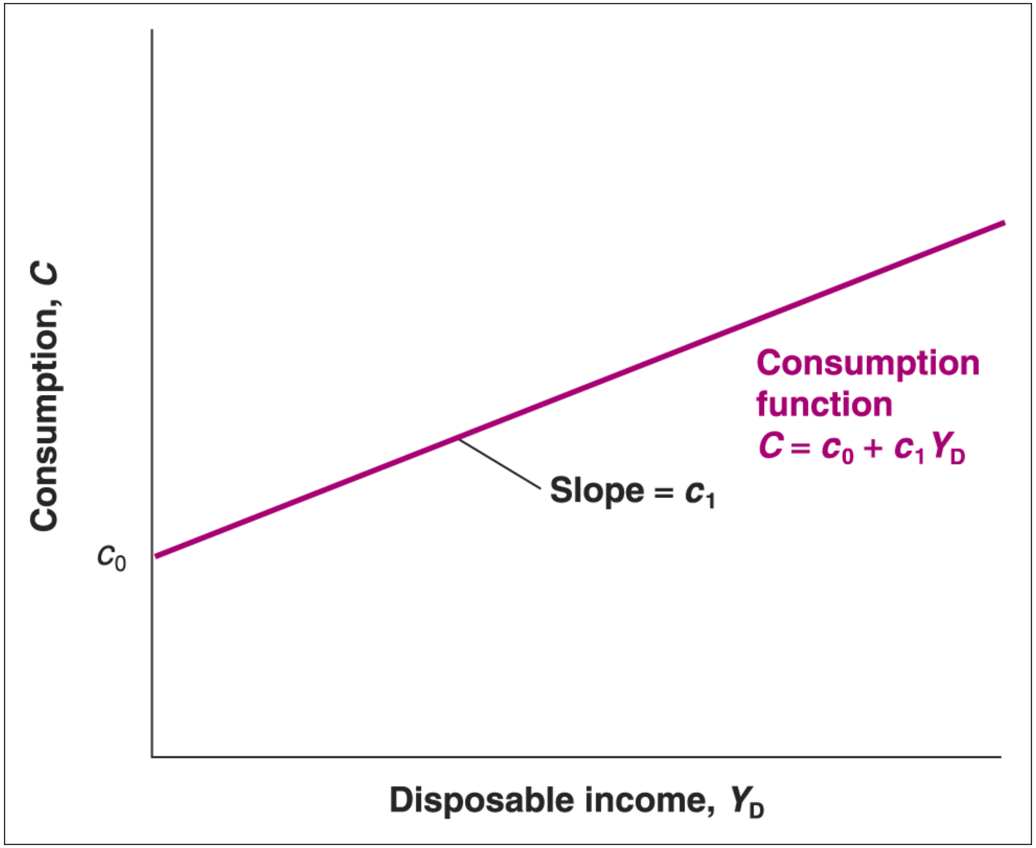
- c_1 is called the **marginal propensity to consume**
  - measures effect of 1 additional euro of disposable income on consumption
  - slope of the consumption function in (C, Y_D) plan
  - $0 < c_1 < 1$ WHY?

- c_0 is called **autonomous consumption**
  - measures the consumption when disposable income is zero
  - captures changes in consumption at given income Y_D
  - intercept of the consumption function in (C, Y_D) plan
  - $c_0 > 0$ WHY?

$$C = C(Y_D)$$

$$Y_D \equiv Y - T$$

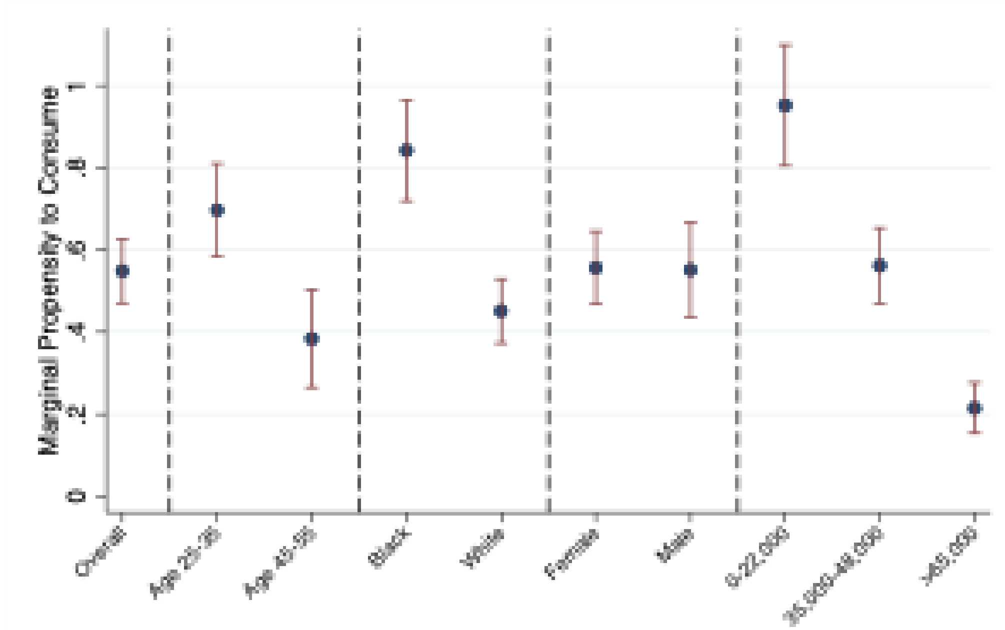
$$C = c_0 + c_1(Y - T)$$



Consumption increases with disposable income but less than 1 for 1

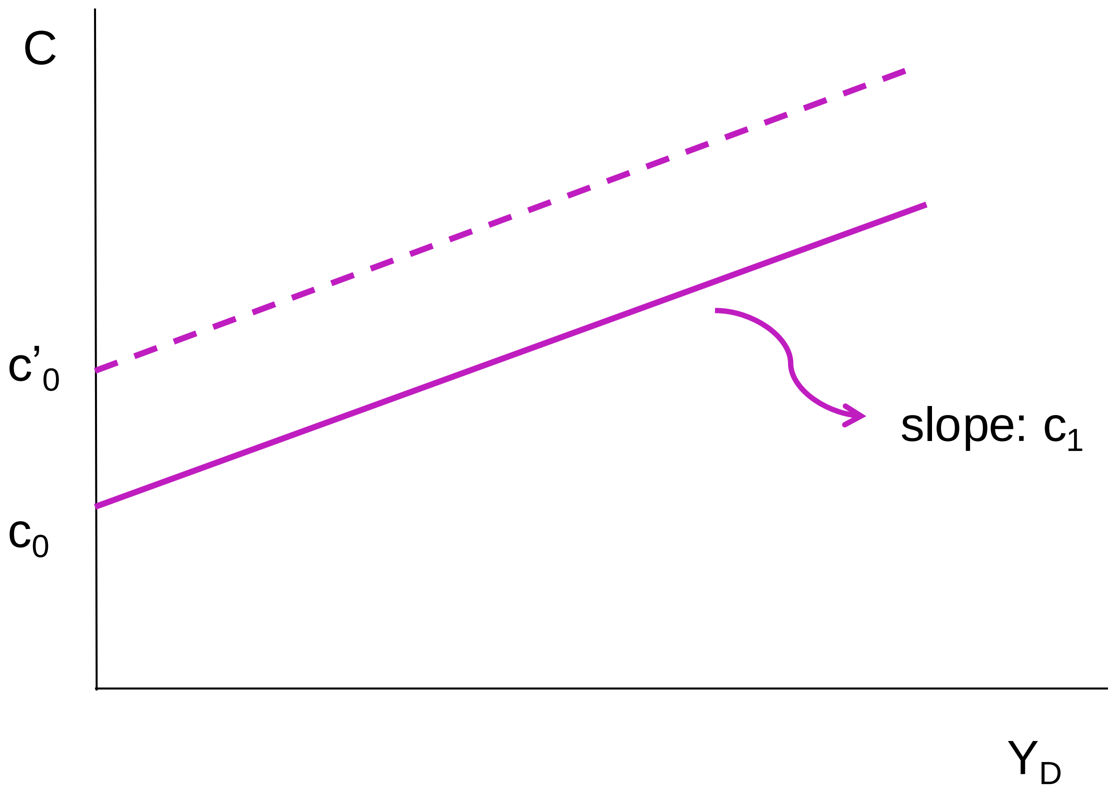
Aside: Heterogeneity in Empirical MPCs

Figure 3: Heterogeneity in Marginal Propensity to Consume Estimates



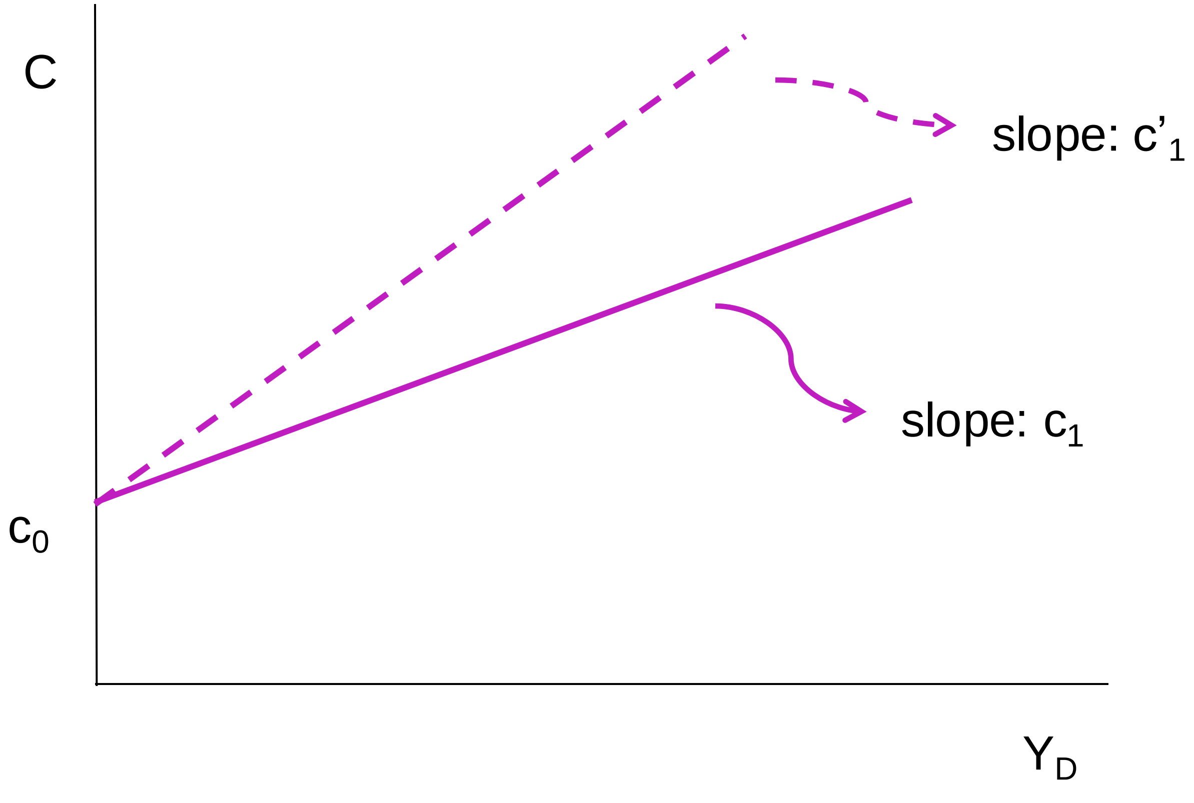
Consumption function I

$c_0 \longrightarrow c'_0 > c_0$



Consumption function II

$$c_1 \longrightarrow c'_1 > c_1$$



Variables that depend on other variables within the model are called **endogenous**

Variables that are not explained within the model are called **exogenous**

Investment (I)

- **Investment (I):** for now investment is taken as given and assumed to be exogenous

$$I = \bar{I}$$

- Later we will endogenize investment

Government spending (G)

- **Government spending (G):** government spending G together with taxes T describes **fiscal policy**, the choice of taxes and spending by the government

- We assume that **G** and **T** are **exogenous** for two reasons:

  - Governments do not behave with the same regularity as consumers or firms → more difficult to write a behavioral equation

  - Macroeconomists want to think about the implications for the economy of alternative government spending and tax decisions → easier to take fiscal policy as exogenous

Equilibrium in the goods market

- **Demand, Z**: sum of consumption, investment and government spending

$$Z \equiv C + I + G$$

$$Z = c_0 + c_1 (Y - T) + \bar{I} + G$$

- **Supply, Y**: firms willing to supply any amount of goods demanded at given prices P

- **Equilibrium**: demand of goods Z must equal supply of goods Y (abstracting from inventories)

$$Y = Z$$

- This is an equilibrium equation

- Demand in turn depend on income which is equal to production

$$Y = c_0 + c_1(Y - T) + \bar{I} + G$$

- Note that Y denotes both production and income

Solving a model

- Solving a model means understanding what factors determine the variables of interest of the model, that is, the endogenous variables

  - algebraically: express the endogenous variables as a function of parameters and exogenous variables

  - graphically: use a graphical analysis to describe how the endogenous variables change when parameters or exogenous variables change

- Three tools

  1. algebra to make sure the logic is correct and to provide quantitative answers
  2. graphs to build the intuition and to provide qualitative answers
  3. words to explain the results and the economic intuition

Solving for equilibrium Y

Using algebra

- The equilibrium equation is

$$Y = c_0 + c_1(Y - T) + \bar{I} + G$$

- Start solving for Y

$$(1 - c_1)Y = c_0 - c_1 T + \bar{I} + G$$

- Finally

$$Y = \frac{1}{1 - c_1}\left(c_0 + \bar{I} + G - c_1 T\right)$$

$$Y = \underbrace{\frac{1}{1-c_1}}_{\text{Multiplier}} \underbrace{\left(c_0 + \bar{I} + G - c_1 T\right)}_{\text{Autonomous spending}}$$

- The term $[c_0 + \bar{I} + \bar{G} - c_1 T]$ is called **autonomous spending**; in fact it is the part of the demand for goods that does not depend on output

- The term 1/(1- c_1) is called the **multiplier**; in fact, it is a number greater than 1 because the propensity to consume c_1 is between 0 and 1

Using graphs

We represent the equilibrium in the (demand Z, income Y) or equivalently in the (output Y, income Y) plan

To find the equilibrium we plot **2 curves**

- Demand curve (ZZ): demand as a function of income

$$Z = \left(c_0 + \bar{I} + G - c_1 T\right) + c_1 Y$$

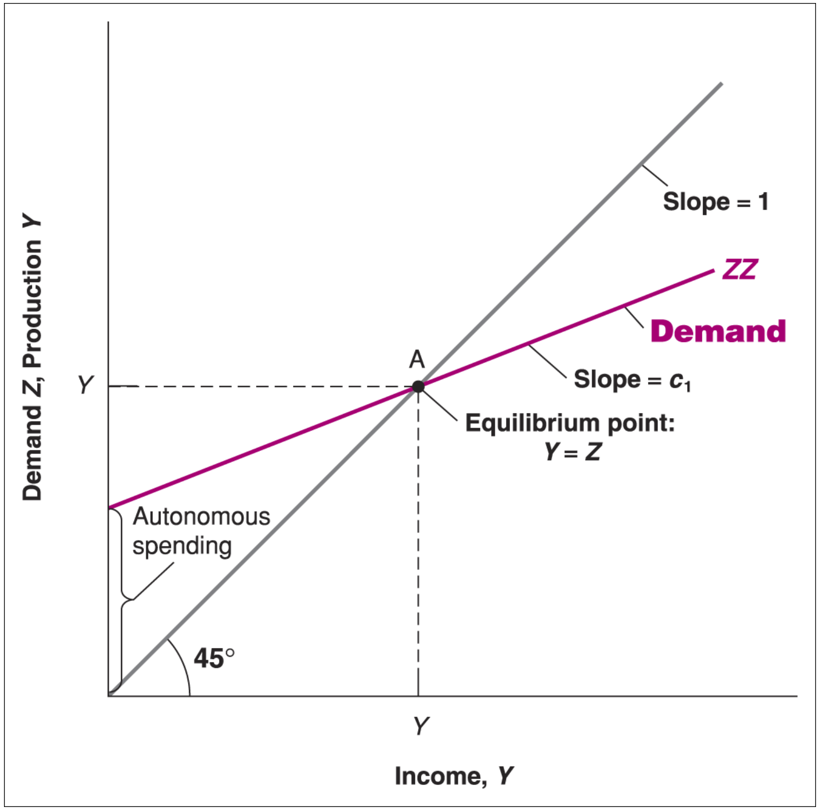
- Supply curve: supply or production as a function of income; since production equals income, this is just the 45° degree line

- First, plot production as a function of income

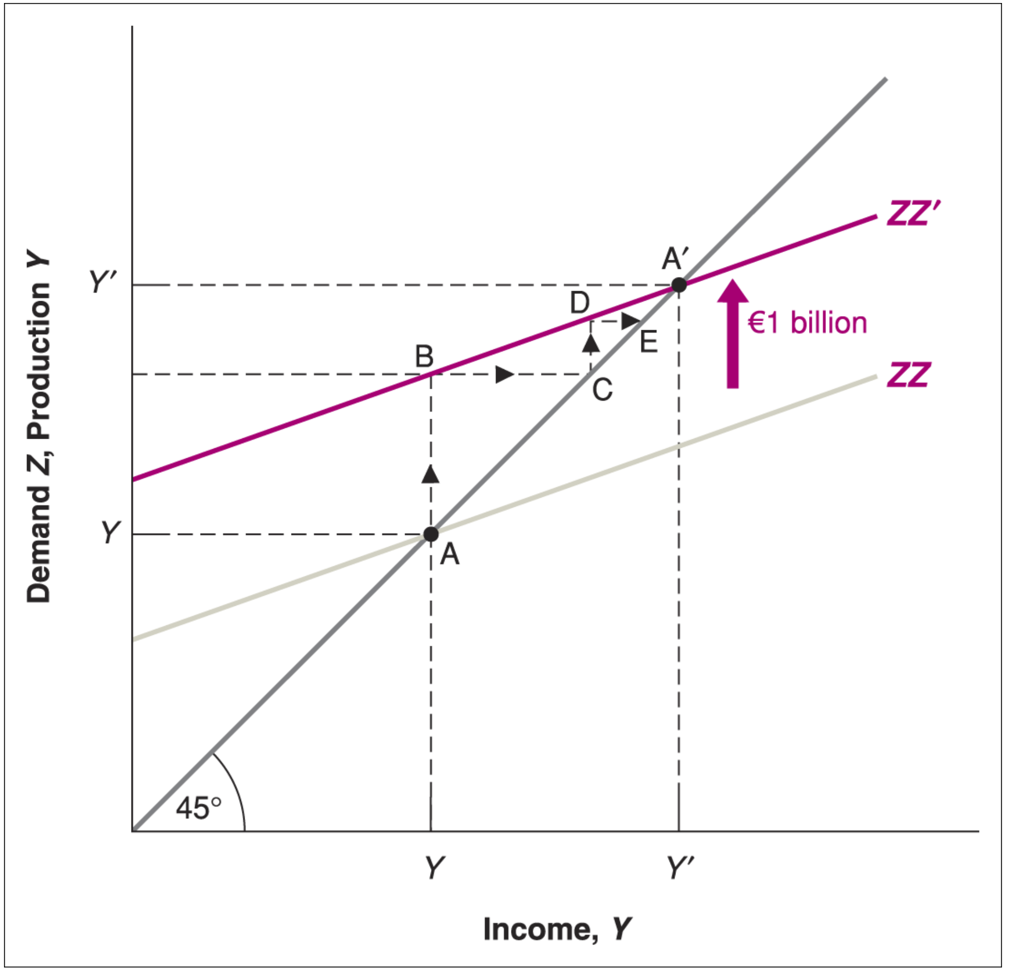
- Second, plot demand as a function of income

$$Z = \left(c_0 + \bar{I} + G - c_1 T\right) + c_1 Y$$

- In equilibrium, production equals demand



Effect of an increase in autonomous spending

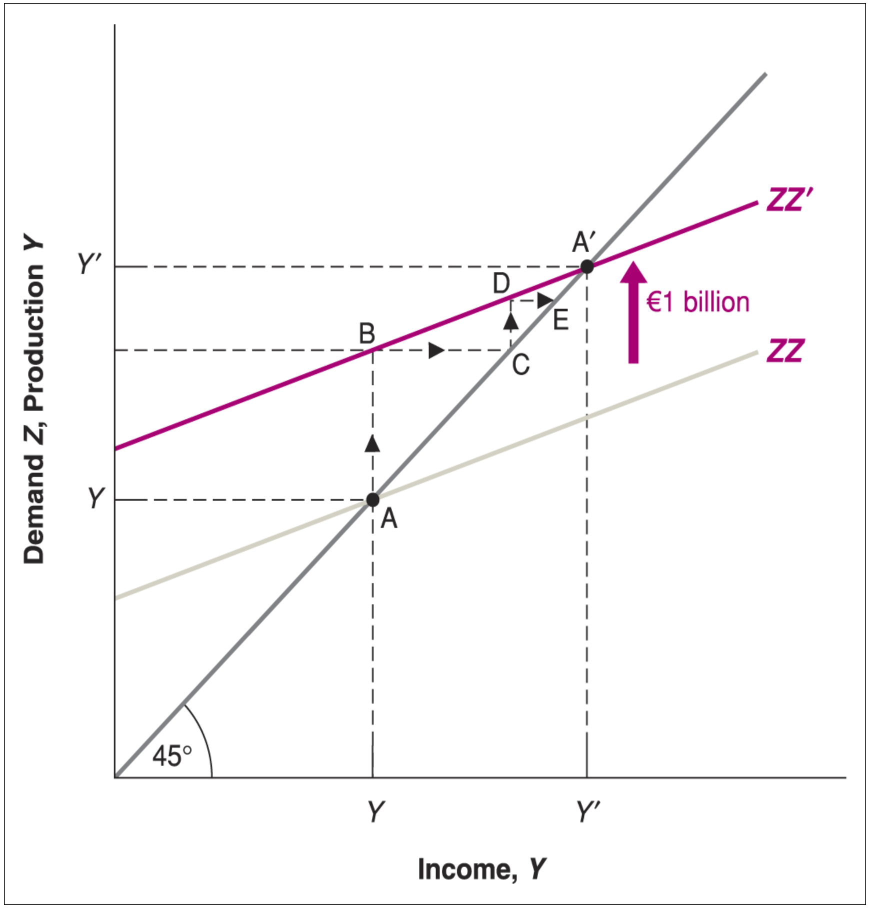


- The first-round increase in demand, represented by the distance *AB,* equals \$1 billion

- This first-round increase in demand leads to an equal increase in production, or \$1 billion, which is also represented by the distance *AB*

- This first-round increase in production leads to an equal increase in income, represented by the distance *BC* and equal to \$1 billion

- Why we do not stop in C? What's wrong with point C?



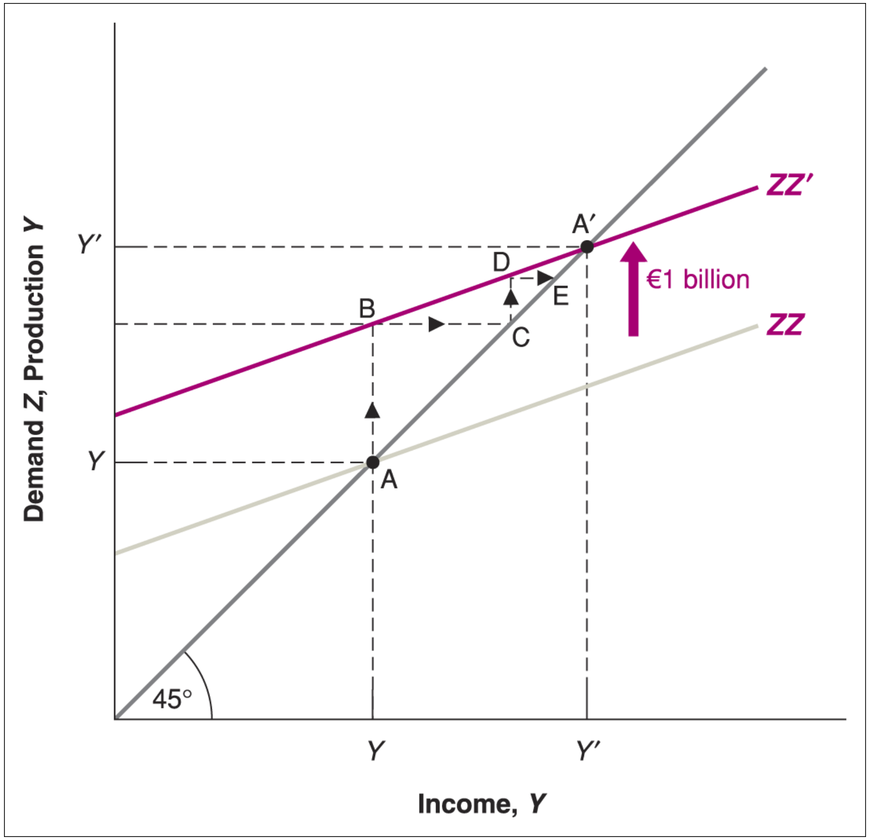
- In C, consumers demand more because income has gone up so that there is a second-round increase in demand

- The second-round increase in demand, represented by the distance *CD*, equals $1 billion times the propensity to consume, that is, c_1 x 1 billion

- This second-round increase in demand leads to an equal increase in production, also represented by the distance *CD*, and thus an equal increase in income, represented by the distance *DE*

- Why we do not stop in E? Because there is a third-round increase in demand

- The third-round increase in demand equals c_1 x (c_1 x 1 billion)….and so on until A'



- By summing the subsequent increases in production, we can compute the total effect on output

- Each subsequent increase in production becomes smaller, forming a geometric series with a common ratio c_1 (where $|c_1|<1$).

- We can then apply the formula for the sum of an infinite geometric series

$$
\begin{aligned}
\text{Change in Y} &= (1 \text{ billion}) + c_1 \times (1 \text{ billion}) + c_1 \times c_1 \times (1 \text{ billion}) + c_1 \times c_1 \times c_1 \times (1 \text{ billion}) + \ldots \\
&= (1 \text{ billion}) \times (1 + c_1 + c_1^2 + c_1^3 + \ldots) \\
&= (1 \text{ billion}) \times \frac{1}{1 - c_1}
\end{aligned}
$$

- The change in Y equals the **initial change in autonomous spending** times the **multiplier**!

Example

$$c_0 = 100, \; c_1 = 0.5, \; \bar{I} = 1000, \; G = 500, \; T = 0$$

Use the formula we found before:

$$Y = \frac{c_0 + \bar{I} + G - c_1 T}{1 - c_1} = \frac{100 + 1000 + 500 - 0}{0.5} = 3200$$

What happens if government spending goes up?

$$c_0 = 100,\ c_1 = 0.5,\ \bar{I} = 1000,\ G' = 1000,\ T = 0$$

Therefore,

$$Y' = \frac{c_0 + \bar{I} + G' - c_1 T}{1 - c_1} = \frac{100 + 1000 + 1000 - 0}{0.5} = 4200$$

So Y goes up! And it goes up by 1000

- Check the numbers: G went up by 500 and c_1 is 0.5

- Thus, Y went up by

$$500 \times \frac{1}{1-0.5} = 1000$$

- Which is the number we found before!

In words

- An increase in demand leads to an increase in production and a corresponding increase in income; this increase in income leads to a further increase in demand, which leads to a further increase in production and so on… subsequent increases in demand and output are smaller and smaller because part of the additional increase in income is saved

- End result: increase in output that is larger than the initial shift in demand, by a factor equal to the multiplier

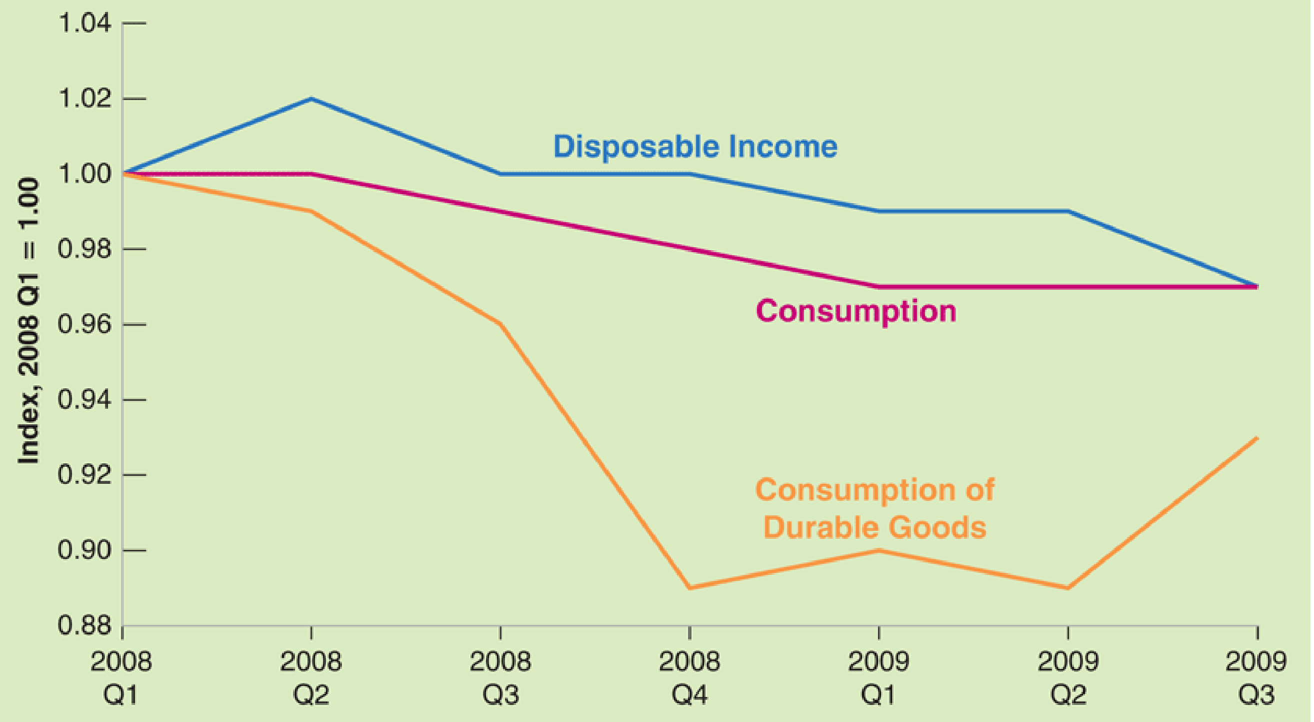
- The value of the multiplier depends on the value of the propensity to consume; to estimate behavioral equations and their parameters, economists use econometrics - a set of statistical methods used in economics

FOCUS: The Lehman bankruptcy, fears of another Great Depression, and shifts in the consumption function

- When people start worrying about the future, they decide to save more even if their current income has not changed
- News about Lehman Brothers going bankrupt in September 2008 reminded people of the Great Depression, as confirmed by the number of searches for "Great Depression" in Google
- Consumption fell even if disposable income had not yet changed

FOCUS: The Lehman bankruptcy, fears of another Great Depression, and shifts in the consumption function

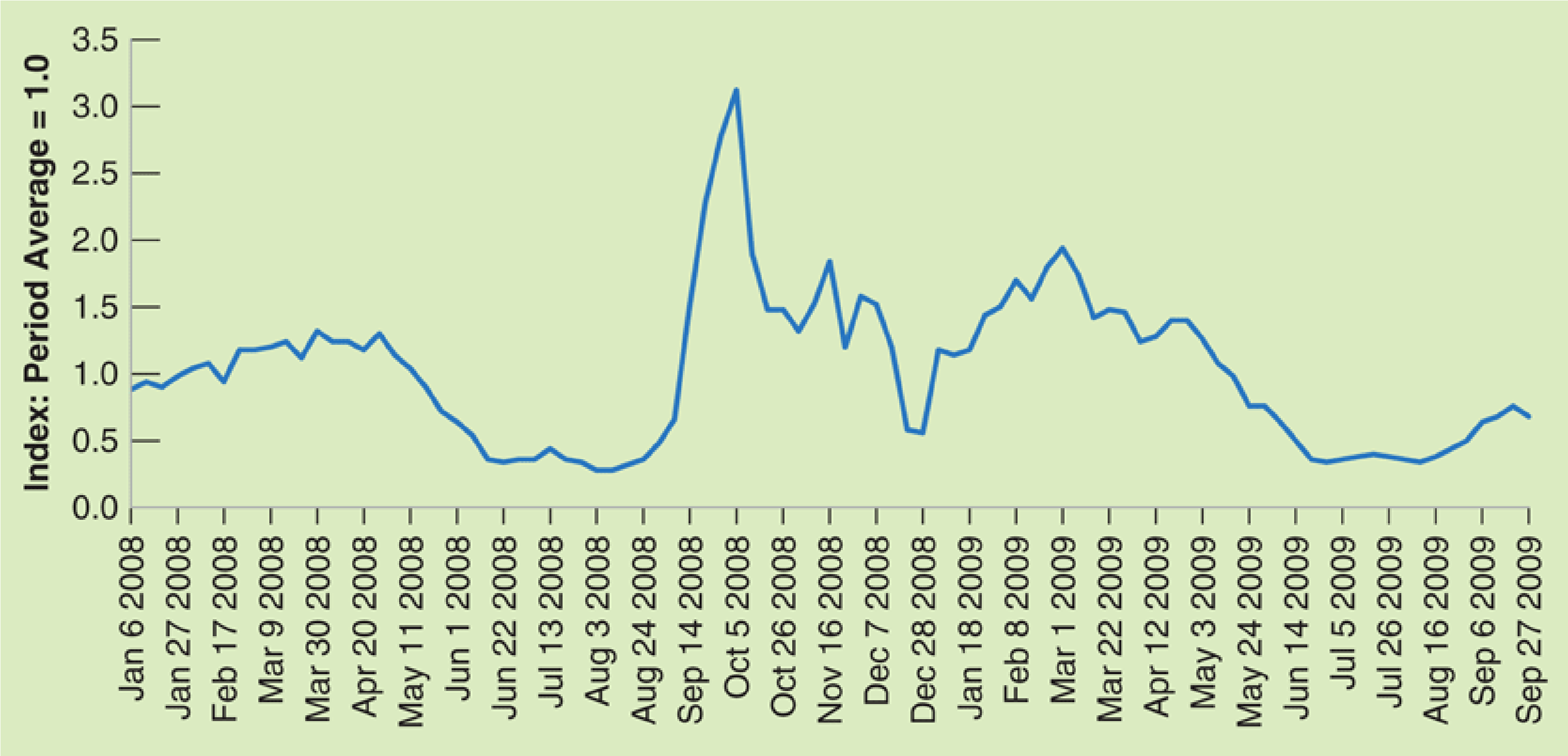
Disposable income, consumption, and consumption of durables in the US, 2008:1- 2009:3



Source: Calculated using series DPIC96, PCECC96, PCDGCC96: Federal Reserve Economic Data (FRED) http://research.stlouisfed.org/fred2/

FOCUS: The Lehman bankruptcy, fears of another Great Depression, and shifts in the consumption function

Google search volume for “Great Depression,” January 2008 to September 2009



Source: Google Trends, “Great Depression.”

Investment = Saving:
An Alternative Way of Thinking About the Goods-Market Equilibrium

Saving is the sum of **private savings** plus **public saving**

- **Private saving** (**S**), is saving by consumers

$$S \equiv Y_D - C \quad \rightarrow \quad S \equiv Y - T - C$$

- **Public saving** equals taxes minus government spending

Public savings = $T - G$

- If $T > G$, the government is running a **budget surplus**—public saving is positive
- If $T < G$, the government is running a **budget deficit**—public saving is negative

$\boxed{\text{Production = Demand}} \longrightarrow Y = C + \bar{I} + G \longrightarrow Y - C - T = \bar{I} + G - T$

$\downarrow$

$\boxed{\text{Investment = Total savings}} \longleftarrow \bar{I} = S + (T - G) \longleftarrow S = \bar{I} + G - T$

$\downarrow$

This explains why the equilibrium condition for the goods market is called the **IS** relation

Put differently, there are 2 ways of stating the condition for equilibrium in goods markets

Production = Demand
Investment = Total savings

Solving for the equilibrium using I = S

- First, what determines savings?

$$S = Y - T - C$$

$$S = Y - T - c_0 - c_1(Y - T)$$

$$S = -c_0 + (1 - c_1)(Y - T)$$

where $(1-c_1)$ is the **marginal propensity to save**

- Thus: **saving** and **consumption decisions** are one and the same: given Y_D, when consumers have chosen C, they have also chosen S

- In equilibrium

$$\begin{aligned}\bar{I} &= S + (T - G) \\ &= -c_0 + (1 - c_1)(Y - T) + (T - G)\end{aligned}$$

- Rearranging we obtain the same expression for equilibrium output

$$Y = \frac{1}{1 - c_1}\left[c_0 + \bar{I} + G - c_1 T\right]$$

The paradox of saving (or the paradox of thrift)

- The paradox of saving is that as people attempt to save more, the result is both a decline in output and unchanged saving
- Suppose at any given level of Y, people save more, that is, they reduce c_0; this increases savings
- The corresponding decrease in consumption reduces equilibrium output and income; this effect decreases savings

$$S = \underbrace{-c_0}_{\uparrow} + \underbrace{(1-c_1)(Y-T)}_{\downarrow}$$

- What is the net effect? We can show that savings do not change since neither investment, nor taxes nor public spending change

$$S = \bar{I} - (T - G)$$

Is the government omnipotent? A warning

Expression for equilibrium output implies that the government can choose G or T to achieve the level of Y it wants. But there are many aspects of reality we have not incorporated into the model:

- Changing government spending or taxes is not always easy
- Responses of consumption, investment, imports, … are hard to assess with much certainty
- Anticipations are likely to matter
- Achieving a given level of output can come with unpleasant side effects in medium run
- Budget deficits and public debt may have adverse implications in long run